



Figure 1: Qualities of NAS solutions (Image source: googleimages.com)

or replacing the existing disks with ones that have higher storage capacity (also called 'scaling up'). The other method involves adding nodes with 'scale out' capacity. Whenever you add hardware, you increase the whole system's performance as well as its capacity.

Performance: This is what we look for while choosing any storage solution. One cannot afford to compromise on the performance of any storage solution, as this may directly impact the performance of the application that uses the given storage solution. Flexible scalability allows users to increase the capacity and performance independently as per their needs and budget.

Reliability: We all look for resources that can be relied upon for a long period of time, and this is the case even when searching for a storage solution.

Affordability: Since budget and pricing are important, an open source storage solution is a good option because it is available free of cost. This is an important factor for small businesses that cannot afford to spend much just for storage solutions.

Availability: Sometimes, data stored in a storage solution is not available when being fetched by any application. This can occur because of some disk failure. We all want to avoid such circumstances, which may lead to unavailability of data. Data should be easily available when it's being accessed.

Simplicity: Even the most advanced storage solutions come with management interfaces that are as good as or better than the traditional storage units. All such interfaces show details about each node, capacity allocation, alerts, overall performance, etc. This is a significant factor to be considered while choosing a storage solution.

Support: Last but not the least, there should be support from the manufacturer or from a group of developers, including the support for applications. Support is quite essential if you plan on installing your database, virtual server farm, email or other critical information on the storage solution. You must make sure that the manufacturer offers the level of support you require.

Some of the available open source storage solutions

Here's a glance at some of the good open source solutions available.

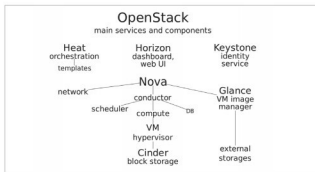


Figure 2: Main services and components of OpenStack (Image source: googleimages.com)

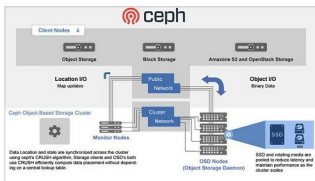


Figure 3: Architecture for the Ceph storage solution (Image source: googleimages.com)

OpenStack: OpenStack is basically a cloud operating system which controls large pools of networking resources, computation and storage throughout a data centre, all of which are managed using a dashboard that gives its administrators the controls while empowering users to provision the resources through a Web interface. The OpenStack Object Storage service helps in providing software that stores and retrieves data over HTTP. Objects (also referred to as blobs of data) are stored in an organisational hierarchy which offers anonymous read-only access or ACL defined access, or even a temporary access. This type of object storage supports multiple token-based authentication mechanisms that are implemented via middleware.

Ceph: This is a type of distributed object storage and file system designed to provide high performance, scalability and reliability. It is built on the Reliable Autonomic Distributed Object Store, and allows enterprises to build their own economic storage devices using different commodity hardware. It is maintained by Red Hat after its acquisition of Inktank in April 2014. It is capable of storing blocks, files and objects as well. It is scale-out, which means that multiple Ceph storage nodes are present on a single storage system which easily handles many petabytes of memory, and simultaneously increases performance and capacity. Ceph has many of the basic enterprise storage features, which include replication, thin provisioning, snapshots, auto-tiering and self-healing capabilities.

RockStar: This is a free and open source NAS solution. The Personal Cloud Server present in it is a very powerful